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Rigid Body Research under Prof. Baruh

Experiment #2: Effect of Drop Height and Inflation Pressure on the Coefficient of Restitution of a Basketball

Model used: Rigid-Body Tracker V6-2

June 2026

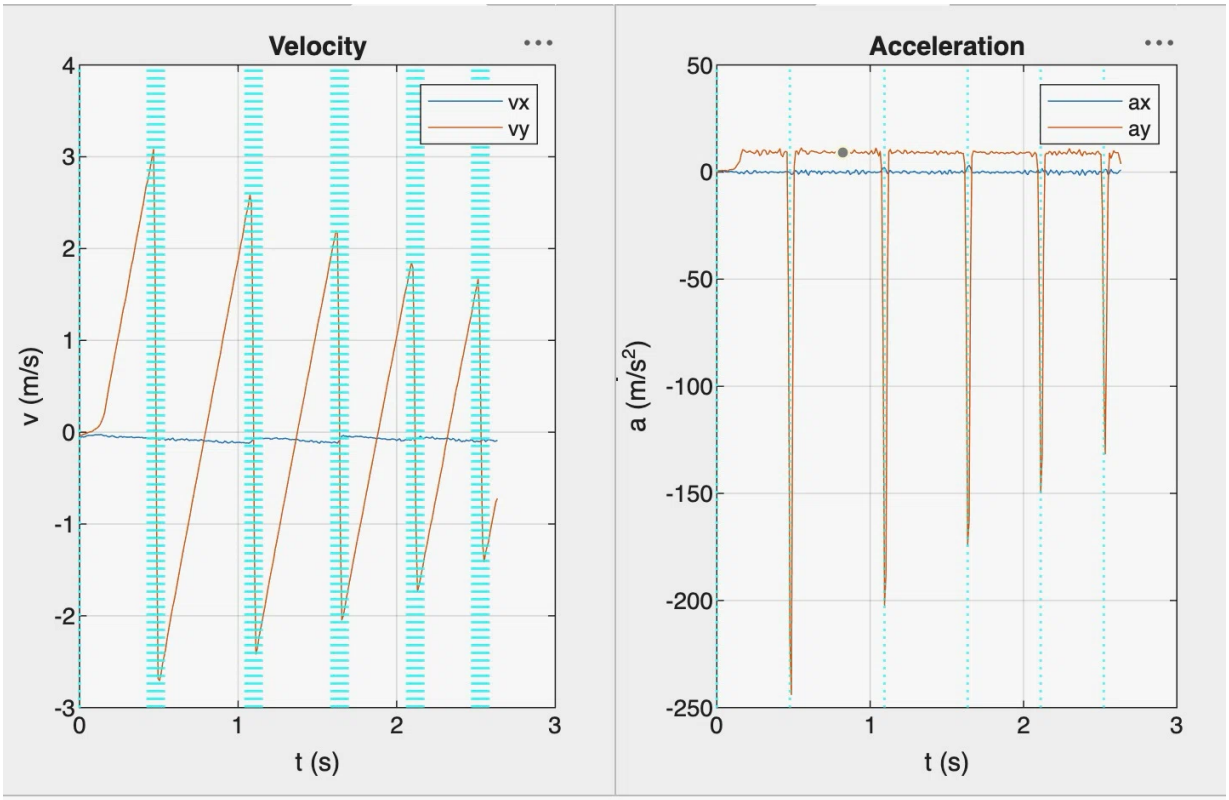
Introduction

This experiment investigated how the coefficient of restitution of a basketball changes with inflation pressure and drop height. The main objective was to determine whether the coefficient of restitution remains approximately constant for different impact velocities and whether inflation pressure significantly affects the bouncing behavior of the ball.

Methods

A basketball was dropped vertically onto a hard floor from heights of 0.5 m, 1.0 m, and 1.5 m. Three inflation pressures were tested: 3.5 PSI, 7.0 PSI, and 10.5 PSI. Two trials were performed for each height-pressure combination, resulting in eighteen total trials.

Data was analyzed using my rigid-body tracking code. The tracker produced very clean curves



with clearly identifiable collision events, allowing automated extraction of e_1, e_2, and e_3 for each trial. To preserve impact information, all coefficient of restitution calculations were performed on unsmoothed data. Below is an example of the type of data used to compute e

Trials Averaged

Height (m)	Pressure (PSI)	e1	e2	e3	avg e	e2/e1	e3/e2
0.5	3.5	0.8565	0.85	0.8535	0.853	0.992	1.004
1	3.5	0.769	0.83	0.815	0.805	1.079	0.982
1.5	3.5	0.811	0.792	0.807	0.803	0.977	1.019
0.5	7	0.867	0.884	0.823	0.858	1.02	0.931
1	7	0.857	0.844	0.834	0.845	0.985	0.988
1.5	7	0.825	0.872	0.839	0.845	1.057	0.962
0.5	10.5	0.935	0.952	0.937	0.941	1.018	0.984
1	10.5	0.879	0.854	0.869	0.867	0.972	1.018
1.5	10.5	0.88	0.873	0.911	0.888	0.992	1.044

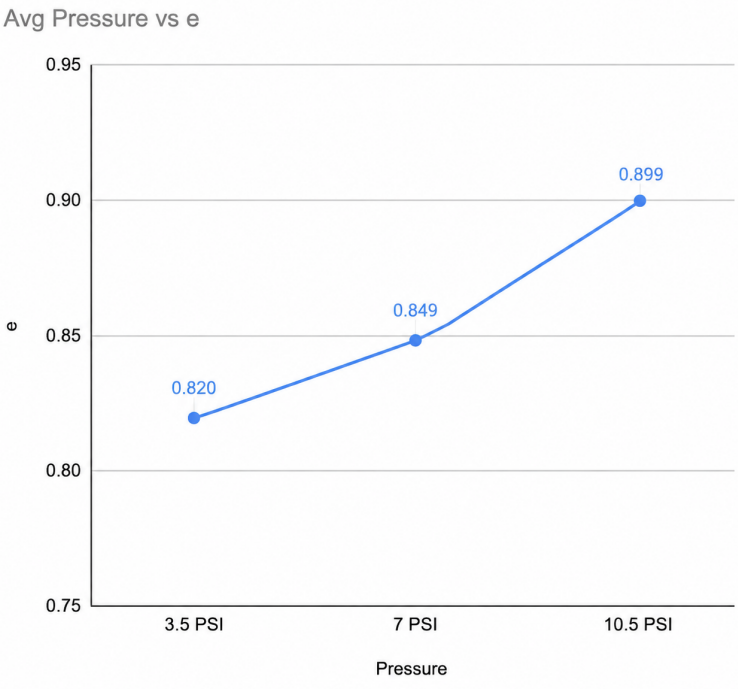
Avg e by pressure:

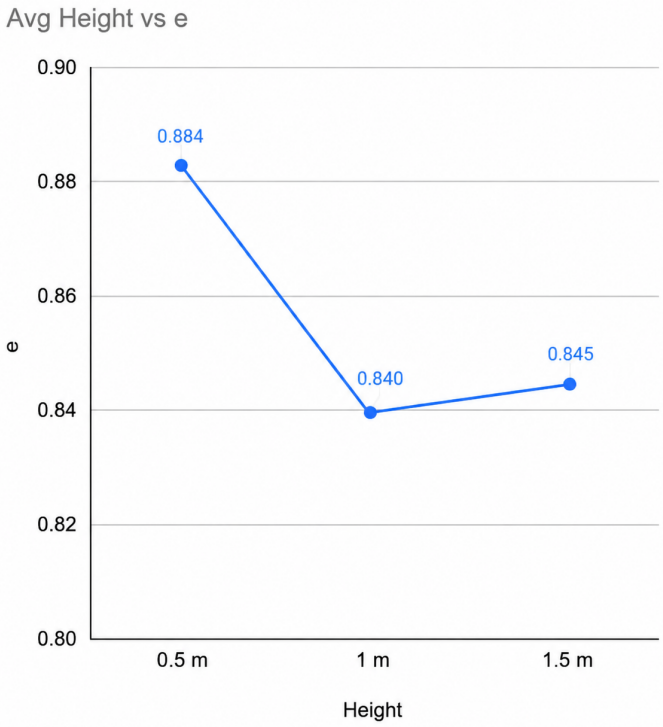
Pressure	e
3.5 PSI	0.82
7 PSI	0.849
10.5 PSI	0.899

Avg e by height:

Height	e
0.5 m	0.884
1 m	0.84
1.5 m	0.845

e2/e1	e3/e2
Average ≈ 1.01	**Average ≈ 0.99**





for the first 3 bounces, completely unsmoothed:

Calculations were done automatically by the code instead of with a physical calculator to reduce experimental error. Each video was recorded at 120 FPS to reduce motion blur and minimize timing uncertainty during collisions. Each of the two trials for each height and pressure value were then averaged.

Results

After averaging the trials, several trends became clear. The coefficient of restitution increased with inflation pressure, as seen clearly by the graph comparing pressure vs e.

The effect of drop height was less noticeable. The average coefficient of restitution was highest at 0.5 m and slightly lower at 1.0 m and 1.5 m. While this suggests a small decrease in coefficient of restitution with increasing height, the trend was weak and no strong relationship was evident within the tested range.

The ratios e_2/e_1 and e_3/e_2 remained close to unity, with average values of approximately 1.01 and 0.99. This suggests that the coefficient of restitution remained approximately constant between successive bounces.

Experimental Error

There were a few sources of experimental error for this experiment. The basketball did not always impact exactly the same location on the floor, introducing small variations in the collision conditions. A small amount of perspective distortion was also observed during recording, especially for the 1.5 m drop. Some energy was likely transferred into rotational motion of the basketball rather than purely translational motion. Only two trials were performed for each test condition. Despite these limitations, the overall trend of increasing coefficient of restitution with increasing inflation pressure remained clear throughout the experiment.